

Test Plan And Strategy

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Date: 12 March 2026

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Test Plan and Strategy

The purpose of this testing project is to verify that the SM Solutions Help Desk System works correctly and meets user requirements before deployment. Testing will focus on the main system features including user registration, login, ticket management, email notifications, and role-based access.

Testing Objectives

- To Ensure user registration and login functionality works correctly
- Verify tickets can be created, assigned, and updated
- Confirm email notifications and alerts are sent correctly
- Test role-based access control for Admin, Support Agent, and User
- Identify bugs, usability issues, and system performance problems
- Provide clear documentation of all findings

Scope of Testing

These following features will be tested:

- User Registration
- User Login / Logout
- Ticket Creation
- Ticket Assignment
- Ticket Status Updates
- Email Notifications
- Role-Based Access Control (Client, Support Staff, Admin)
- System Alerts

Test Strategy

The following testing approaches will be used:

- Functional Testing: Verify that each feature works according to the system requirements.
- Usability Testing: Check whether the system is easy to use and understand.
- Negative Testing: Testing invalid actions to ensure the system handles them correctly.

- Role based Testing: Testing using different roles such as client, support staff and Admin.
- Performance Testing: Check if the system responds in a reasonable time.

Test Environment

Testing will be performed using the following:

- **Browser:** Google Chrome / Microsoft Edge/ Firefox
- **Operating System:** Windows 10
- **Test System:** SM Solutions Help Desk System (Test Environment)
- **Type of devices:** Desktop, Laptop and Mobile device

Test Data

Sample test data will include:

- Valid and invalid email addresses
- Strong and weak passwords
- Sample support tickets

Defect Reporting Process

All issues found during testing will be documented with the following details:

- Bug ID
- Bug Description
- Steps to Reproduce
- Expected Result
- Actual Result
- Severity Level
- Screenshot Evidence

Test Deliverables

The following documents will be produced:

- Test Plan
- Test Cases
- Bug Reports
- Final Testing Report

Conclusion

Testing will ensure the SM Solutions Help Desk System functions correctly, provides a good user experience, and is ready for deployment. All defects and usability issues will be clearly documented to help developers fix them efficiently.